Fig. 3: Image obtained with command imshow(G)

Fig. 4: Image obtained with imshow(F) command

less than zero is displayed as black. To get the values within this range, a division factor can be used. The larger the division factor, the darker will be the image.

As an example, if you give the command imshow(G), the image shown on the desktop is given in Fig. 3. 'imshow(F)' displays the image shown in Fig. 4. 'imshow(f, [low high])' displays as black all values less than or equal to 'low' and as white all values equal to or greater than 'high.' The values in between are displayed as intermediate intensity values.

'imshow(f, [ ])' sets variable low to the minimum value of array 'f' and high to its maximum value. This helps in improving the contrast of images having a low dynamic range.

The Image tool in the image processing toolbox provides a more interactive environment for viewing and navigating within images, displaying detailed information about pixel values, measuring distances and other useful operations. To start the image tool, use the imtool function. The following statements read the image Penguins_grey.jpg saved

VARIOUS IMAGE TOOLS AND THEIR CAPABILITIES

Image tools	Capabilities
Pixel information	Displays information about the pixel under the mouse pointer
Pixel region	Superimposes pixel values on a zoomed-in pixel view
Measure distance	Measures the distance between two pixels
Image information	Displays information about images and image files
Adjust contrast	Adjusts the contrast of the displayed image
Crop image	Defines a crop region and crops the image
Display range	Shows the display range of the image data
Overview	Shows the currently visible image
Choose colormap	Shows the image under different colour settings

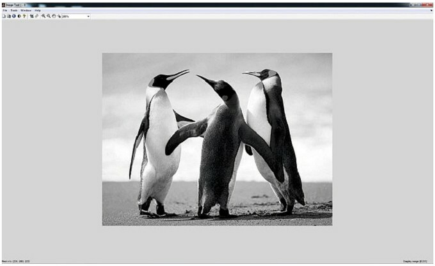

Fig. 5: The window that appears when using the image tool

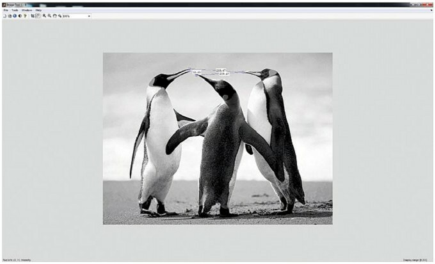

Fig. 6: The Measure Distance tool under Tools tab is used to measure distances between the beaks of different penguins